Problem 3

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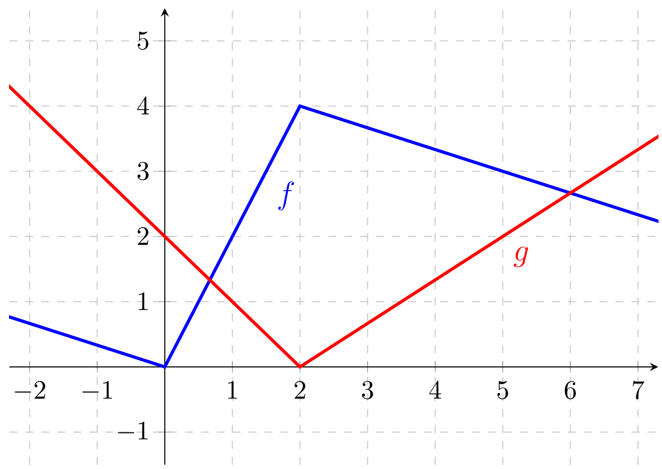
Problem 1 *Given the following graphs of f and g (both piecewise linear functions), define new functions $u(x) = f(g(x))$ and $v(x) = f(x)g(x)$. Find:*



(a) $u'(1)$

Explanation. $u'(x) = \dfrac{d}{dx}(f(g(x))) = f'(g(x)) \cdot g'(x)$. So,

$$\begin{aligned} u'(1) &= f'(g(1)) \cdot g'(1) \\ &= f'(1) \cdot (-1) \\ &= (2)(-1) = -2 \end{aligned}$$

(b) $v'(1)$

Explanation. $v'(x) = \dfrac{d}{dx}(f(x)g(x)) = f'(x)g(x) + f(x)g'(x)$. So,

$$\begin{aligned} v'(1) &= f'(1)g(1) + f(1)g'(1) \\ &= (2)(1) + (2)(-1) \\ &= 0 \end{aligned}$$

(c) $\displaystyle\lim_{x\to 1} \frac{\sqrt{g(x)} - 1}{x - 1}$

Explanation. This is asking for $\left[\frac{d}{dx}(\sqrt{g(x)})\right]_{x=1}$. By the Chain Rule, $\frac{d}{dx}(\sqrt{g(x)}) = \frac{g'(x)}{2\sqrt{g(x)}}$ so

$$\begin{aligned}\left[\frac{d}{dx}(\sqrt{g(x)})\right]_{x=1} &= \frac{g'(1)}{2\sqrt{g(1)}} \\ &= \frac{-1}{2\sqrt{1}} \\ &= -\frac{1}{2}.\end{aligned}$$